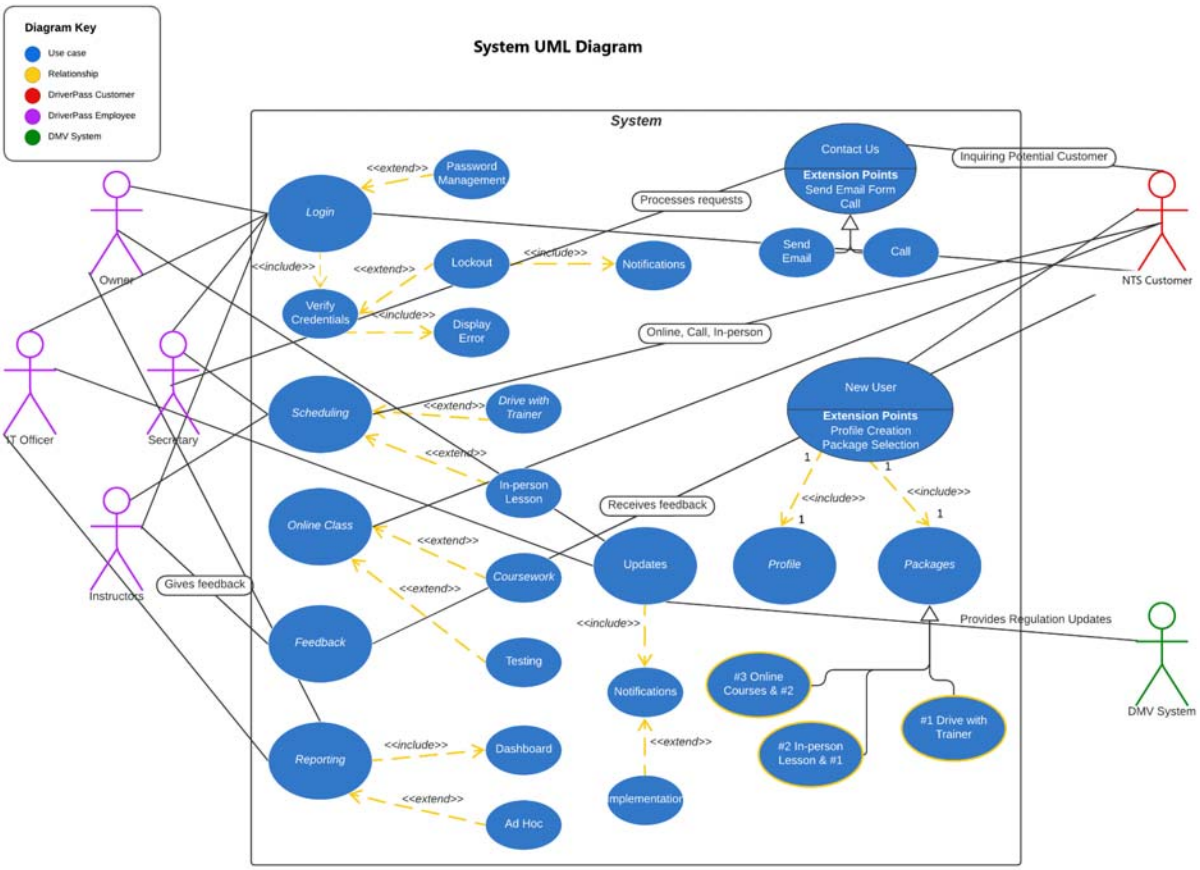
UML and System Designs

Overview

The purpose of this document is to incorporate key visuals regarding the updated New Driver System (NTS) and the technical requirements needed for a successful launch. Included are the UML use case diagram, UML activity diagrams for two key activities, a UML sequence diagram for the DMV system update process, and a UML class diagram for greater detail. Together, these should provide a view of our process in developing the NTS system.

UML Use Case Diagram

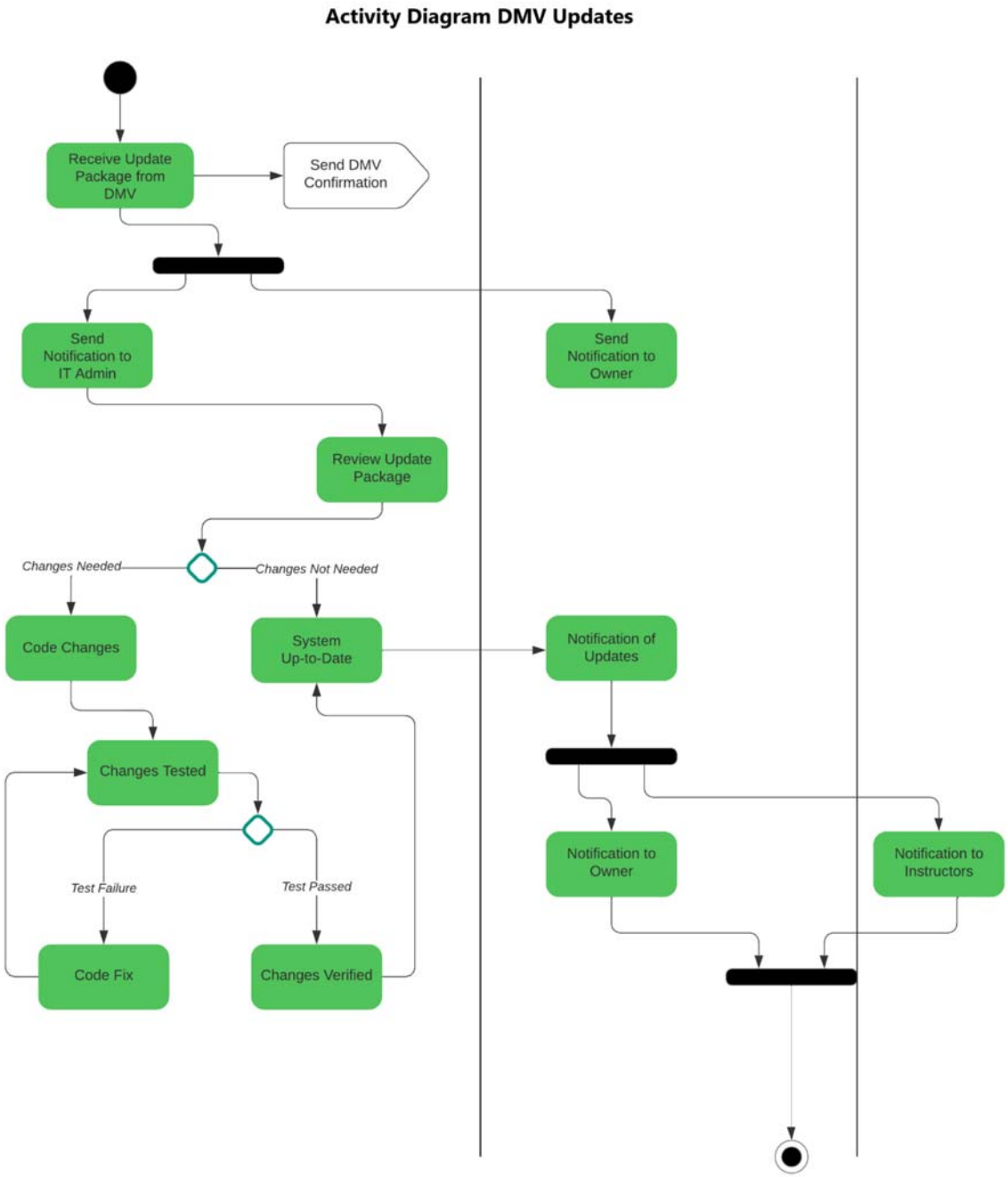
Below is the Use Case Diagram, which is the high-level overview of the entire system, who interacts with it, and the key functions outlined in the requirements gathering meeting. Highlights include the security around logging in, package selection for NTS clients, scheduling of appointments, updates from the DMV, reporting, and coursework and testing. The main users are the owner, IT admin, secretary, instructors, clients, and the DMV system.



UML Activity Diagrams

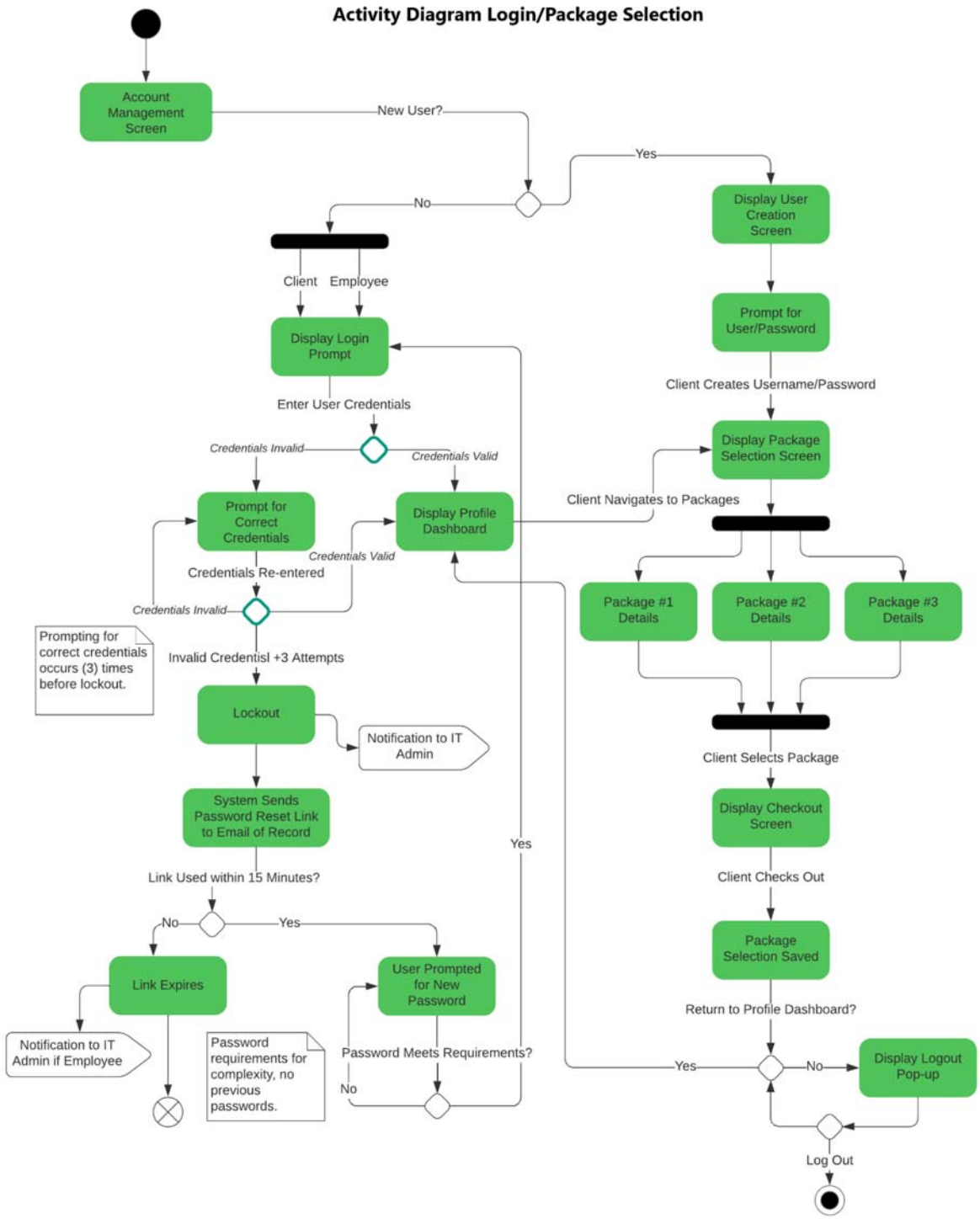
DMV System Updates

The first activity diagram is for the DMV system updates process. Note this process includes verification back to the DMV of receipt of updates and notifications to the IT admin, owner, and instructors.



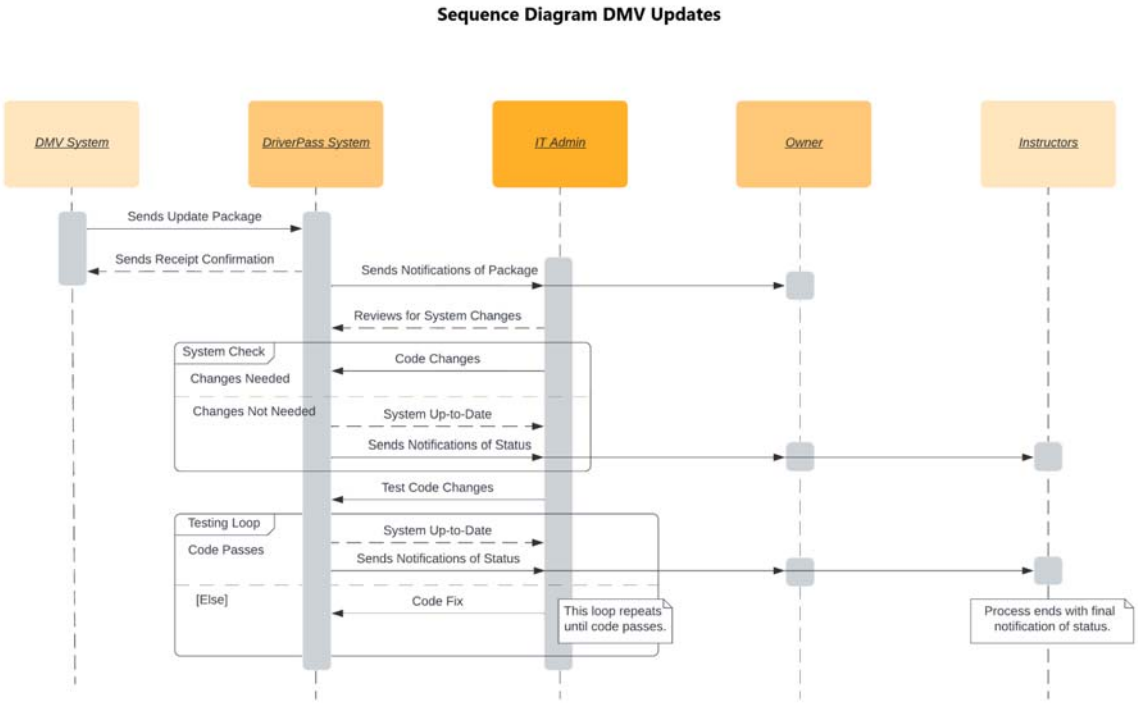
Login/Package Selection

The second activity diagram is for the login/package selection process. These two functions were incorporated together as one cannot select a package without first having access to the NTS system. New users would need to create a profile for access and then select a package. Existing users would need to log in to the system to change a package.



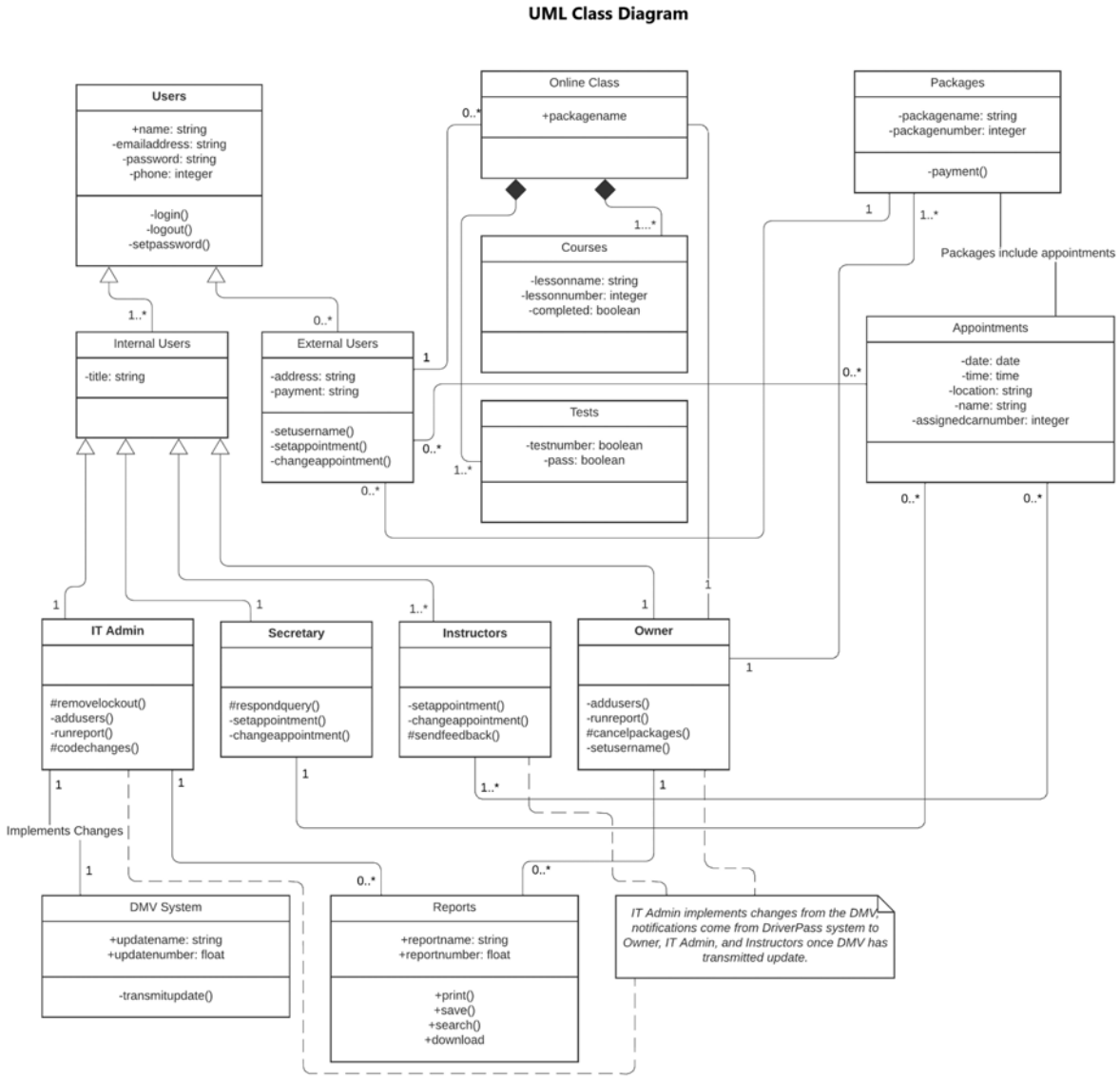
UML Sequence Diagram

Below is the sequence diagram for the DMV Update process. Please take note of the testing loop; this loop will continue until all test failures are remedied.



UML Class Diagram

This is the class diagram, which shows the various key components of the system and how they are related to each other. Take note that the subclasses of the internal users have special actions, they can do that cannot be shared by all users.



Technical Requirements

Focusing on the above diagrams for the DMV updates and login/package selection processes, the following technical requirements were developed for those scenarios.

- Contract/Agreement with DMV for receipt of update packages.
- Staff to make code changes for updates and future maintenance.
- Cloud Computing: Cloud storage, back-up, hosting, and development (f5, 2010, January 10).
- Users' mobile hardware memory requirement of greater than 2GB, 6GB recommended (Amadeo, 2022, September 9).
- Backward compatibility for Android to 10 (Queen Cake) (Android, 2022, September 9) and iOS 15 (Apple, 2022, June 6).
- Users' computer hardware requirements for Windows 10 or Mac OS 11 (Big Sur) or newer.
- All users will need internet access to access NTS.
- Clients must have a credit card/debit card to purchase a subscription.
- The financial link between the system and NTS' business banking account for revenue posting.

Rationale

NTS needs updates from the DMV to keep the system up to date with the latest regulations. Staff are required to make code changes for any updates and potential future maintenance or bug fixes, which are not to be completed by the IT admin. NTS needs to be accessible at any time from anywhere, hence the need for Cloud-based computing. Mobile user RAM recommendation is based on the supported operating system needs. Backward compatibility with the oldest supported operating system increases client accessibility. An internet connection is required to access the system. Users signing up for packages need to make payments with a credit card or debit card, and those transactions must be able to post to the NTS business banking account.

References

Amadeo, R. (2022, September 9). Android 13 Raises Minimum System Requirement to 2GB of RAM, 16GB of Storage. arsTechnica. Retrieved from: https://arstechnica.com/gadgets/2022/09/android-13-raises-minimum-system-requirement-to-2gb-of-ram-16gb-of-storage/

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